

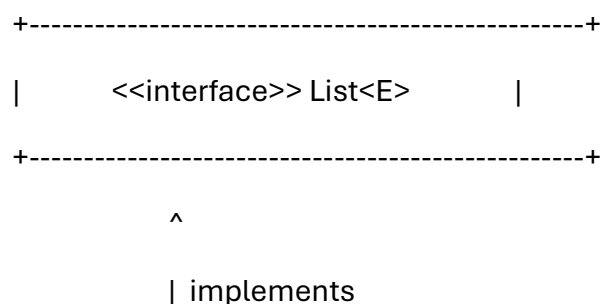
Aim

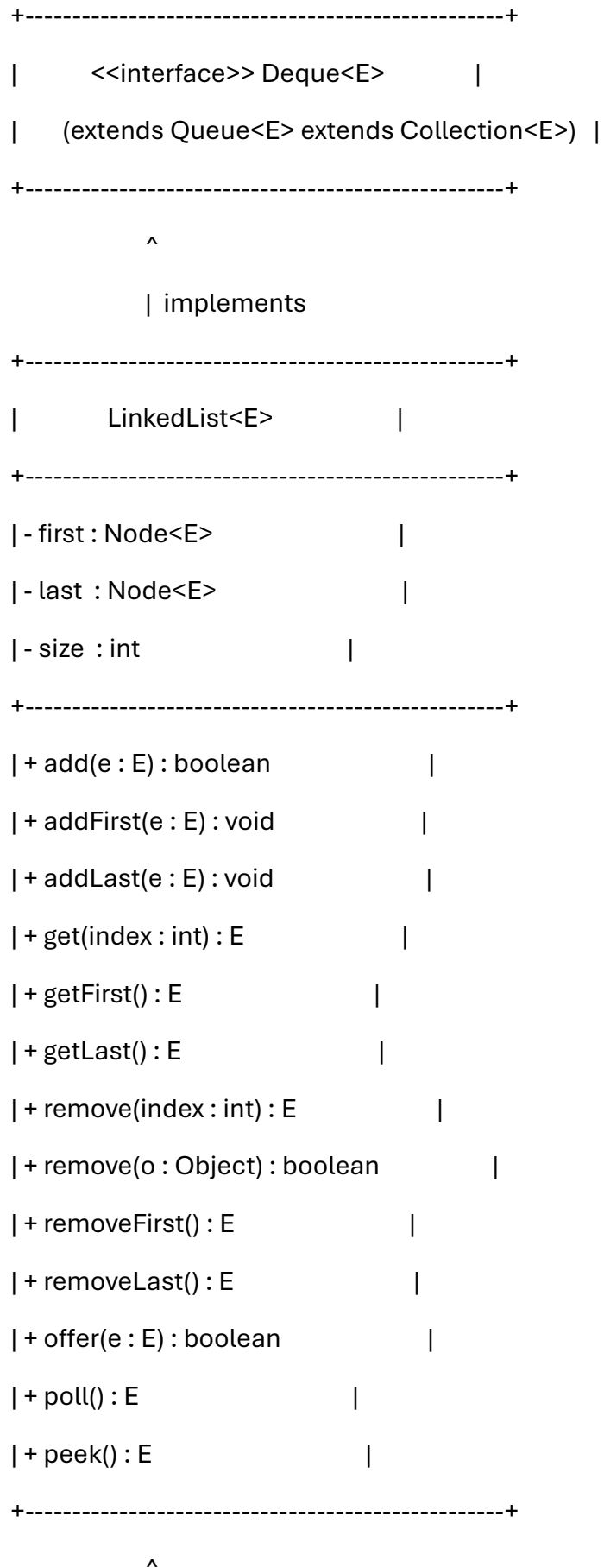
To write a Java program that demonstrates the LinkedList class (a doubly-linked list implementation) using its methods add(), addFirst(), addLast(), get(), getFirst(), getLast(), remove(), removeFirst(), removeLast(), offer(), poll() and peek().

Algorithm

1. Start.
2. Import the java.util.LinkedList class.
3. Create a LinkedList<String> object named list.
4. Use add() to append "Apple", "Banana" and "Cherry" to the end of the list. Display the list.
5. Use addFirst("Mango") to insert at the beginning and addLast("Grapes") to insert at the end. Display the list.
6. Use get(2) to read the element at index 2.
7. Use getFirst() and getLast() to read the first and last elements.
8. Use remove(2) to delete the element at index 2, then remove("Cherry") to delete that element by value. Display the list after each.
9. Use removeFirst() and removeLast() to remove and return the first and last elements. Display the list.
10. Use offer() to add "Orange" and "Papaya" at the tail.
11. Use peek() to view the head element without removing it.
12. Use poll() to retrieve and remove the head element. Display the list.
13. Clear the list and call poll() and peek() to show that both return null on an empty list.
14. Stop.

UML Diagram





| uses

+-----+

| LinkedListDemo |

+-----+

| + main(args : String[]) : void |

+-----+

Program:

```
package Observations;
```

```
import java.util.LinkedList;
```

```
public class LinkedListClass {
```

```
    public static void main(String[] args) {
```

```
        LinkedList<String> list = new LinkedList<>();
```

```
        // add(E e)
```

```
        list.add("Apple");
```

```
        list.add("Banana");
```

```
        list.add("Cherry");
```

```
        System.out.println("After add()        : " + list);
```

```
        // addFirst(E e) / addLast(E e)
```

```
        list.addFirst("Mango");
```

```
        list.addLast("Grapes");
```

```
        System.out.println("After addFirst/addLast : " + list);
```

```
        // get(int index)
```

```
        System.out.println("get(2)        : " + list.get(2));
```

```
// getFirst() / getLast()

System.out.println("getFirst()      : " + list.getFirst());

System.out.println("getLast()       : " + list.getLast());
```

```
// remove(int index) / remove(Object o)

list.remove(2);

System.out.println("After remove(2)   : " + list);

list.remove("Cherry");

System.out.println("After remove(\"Cherry\") : " + list);
```

```
// removeFirst() / removeLast()

System.out.println("removeFirst()     : " + list.removeFirst());

System.out.println("removeLast()      : " + list.removeLast());

System.out.println("After removeFirst/Last : " + list);
```

```
// offer(E e)

list.offer("Orange");

list.offer("Papaya");

System.out.println("After offer()      : " + list);
```

```
// peek()

System.out.println("peek()           : " + list.peek());
```

```
// poll()

System.out.println("poll()           : " + list.poll());

System.out.println("After poll()        : " + list);
```

```

// poll() / peek() on an empty list

list.clear();

System.out.println("poll() on empty list : " + list.poll());

System.out.println("peek() on empty list : " + list.peek());

}

}

```

Output:

```

PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java> & 'C:\Program Files\Java\jdk-11.0.2\bin\java.exe' '-cp' 'C:\Users\Sahithi_pyla\AppData\Roaming\Code\User\workspaceS
bervations\LinkedListClass'
After add() : [Apple, Banana, Cherry]
After addFirst/addLast : [Mango, Apple, Banana, Cherry, Grapes]
get(2) : Banana
getFirst() : Mango
getLast() : Grapes
After remove(2) : [Mango, Apple, Cherry, Grapes]
After remove("Cherry") : [Mango, Apple, Grapes]
removeFirst() : Mango
removeLast() : Grapes
After removeFirst/Last : [Apple]
After offer() : [Apple, Orange, Papaya]
peek() : Apple
poll() : Apple
After poll() : [Orange, Papaya]
poll() on empty list : null
peek() on empty list : null
PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java>

```

Result:

The program was successfully executed and the LinkedList methods add(), addFirst(), addLast(), get(), getFirst(), getLast(), remove(), removeFirst(), removeLast(), offer(), poll() and peek() were demonstrated. poll() and peek() return null on an empty list, whereas getFirst() and removeFirst() would throw NoSuchElementException.